

Miji Jeong | Curriculum Vitae

Korea Astronomy and Space Science Institute (KASI)
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Academic Position

Postdoctoral Researcher, Korea Gemini Office Sep 2025–Present
Korea Astronomy and Space Science Institute (KASI)
Hosts: Dr. Ji Yeon Seok, Dr. Soung-Chul Yang
Support Gemini users with proposal and Phase II preparation, observing strategy, configuration, and reduction guidance.
Review observing plans and work with Gemini staff on technical and operational issues as needed.

Education

Ph.D. Astronomy, Space Science and Geology Mar 2020–Aug 2025
Chungnam National University (CNU), Daejeon, Republic of Korea
Advisor: Dr. Young Sun Lee
Thesis: The Chemical Fingerprints of the Early Milky Way: Insights from Metal-Poor Halo Stars

M.Sc. Astronomy, Space Science and Geology Aug 2017–Feb 2020
CNU, Daejeon, Republic of Korea
Advisor: Dr. Young Sun Lee
Thesis: Chemodynamical Properties of Extremely Metal-Poor Stars

B.Sc. Astronomy and Space Science Mar 2011–Aug 2017
CNU, Daejeon, Republic of Korea

B.A. Philosophy Mar 2011–Aug 2017
CNU, Daejeon, Republic of Korea
Advisor: Dr. Daesuk Han
Bachelor's Thesis: A Philosophical Reflection on Language Games in Wittgenstein's *Philosophical Investigations*

Research Experience

Visiting Student, NSF NOIRLab Feb 10–17, 2024
Tucson, Arizona, USA Jul 10–17, 2024
Collaborator: Dr. Vinicius Placco
Refined line lists and high-resolution abundance analyses for Gemini-S/GHOST and LBT/PEPSI, including r-process and actinide elements; tested analysis choices and interpreted results with collaborators.

Visiting Scholar, University of Notre Dame Sep 2023–Feb 2024
Department of Physics and Astronomy, Indiana, USA
Advisor: Dr. Timothy Beers
Developed high-resolution follow-up plans for chemically peculiar and r-process-enhanced stars, including target selection, LBT/PEPSI observing strategy, proposal preparation, data-quality checks, and abundance analysis.

Student Intern, Gemini South / GHOST Instrument Team Feb–Aug 2023
Gemini Observatory / NSF NOIRLab, La Serena, Chile
Advisor: Dr. Jeong-Eun Heo
Reduced GHOST System Verification data with DRAGONS, compared observed performance with ITC predictions, tested pipeline products, and provided feedback on reduction workflows and user documentation.

Co-Investigator, Korea Aerospace Research Institute (KARI) Jul–Dec 2021
Daejeon, Republic of Korea
Project: Development of Open Source Satellite Image Analysis Tool (KOMPSAT)
Developed analysis tools for KOMPSAT data, prepared user documentation, and provided user training at KARI.

Honors/Awards

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| • Graduate Dean's Award for Excellent Journal Paper | 2023, CNU |
| • Graduate Student Leadership Scholarship | 2021, CNU |
| • Academic Excellence Scholarship | 2017, CNU |

Selected Funded Projects

Co-Investigator | KRW 10 million

Jul-Dec 2026

KASI Mini-Project to Enhance and Disseminate AI Capabilities, Track B

Project: Development and Dissemination of an AI-Assisted Administrative Support System

Professional Service

Academic Service

- Member, Korea Time Allocation Committee, K-GMT Science Program 2026–Present
- Peer Reviewer for *The Astrophysical Journal* 2026–Present
- Session Chair, Stars/Star Clusters/Exoplanets 2, 113rd KAS Spring Meeting Apr 2026
- Student Proposal Reviewer, Gemini Observatory Fast Turnaround 2024B Sep 2024

Leadership and Organizational Roles

- Organizing Committee, 7th K-GMT Science Program Users' Meeting Aug 2026
- Vice Chair of the Organizing Committee, Korea Young Astronomers Meeting Jan–Dec 2021
- Organizing Committee, Korea Young Astronomers Meeting Jan 2018–Dec 2020

Volunteers

- IAUGA 2022 XXXIst General Assembly, International Astronomical Union, Busan, Korea Aug 2–21, 2022
- 8th East-Asia School and Workshop on Laboratory, Space, and Astrophysical Plasmas, Daejeon, Korea Jul 30–Aug 3, 2018
- APRIM 2014: 12th Asia-Pacific IAU Regional Meeting, Daejeon, Korea Aug 19–22, 2014

Observing Experiences

Visitor Mode

- Gemini Observatory, Hawaii, USA
 - 8.1 m Gemini-North Telescope (GN), Gemini Remote Access to CFHT ESPaDOnS Spectrograph (GRACES): 6 nights
- Gemini Observatory, Cerro Pachón, Chile
 - 8.1 m Gemini-South Telescope (GS), Gemini High-resolution Optical Spectrograph (GHOST): 9 nights (System Verification run)
- Bohyunsan Optical Astronomy Observatory, Yeongcheon, Korea
 - 1.8 m Bohyunsan Optical Astronomy Observatory Telescope, Bohyunsan Optical Echelle Spectrograph (BOES): 3.5 nights
- Dukheung Astronomy Observatory, Goheung, Korea
 - 1.0 m Dukheung Astronomy Observatory Telescope, Shelyak eShel Spectrograph: 15 nights

Service Mode

- Gemini Observatory, Hawaii, USA
 - 8.1 m Gemini-North Telescope, GRACES, Gemini Multi-Object Spectrographs (GMOS), M dwarf Advanced Radial velocity Observer Of Neighboring exoplanets (MAROON-X), The Immersion GRating INfrared Spectrograph 2 (IGRINS-2)
- Gemini Observatory, Cerro Pachón, Chile
 - 8.1 m Gemini-South Telescope, Gemini Multi-Object Spectrographs (GMOS)
- Mt Graham International Observatory, Arizona, USA
 - Two 8.4 m Large Binocular Telescope (LBT), Potsdam Echelle Polarimetric and Spectroscopic Instrument (PEPSI)

Remote Mode

- National Astronomical Observatory of Japan, Hawaii, USA
 - 8.2 m Subaru Telescope, High Dispersion Spectrograph (HDS): 2 nights

Successful Observing Proposals as PI

2026

- GS/GHOST | 10.4 hr GS-2026B-Q-108
“Revealing Light-to-Heavy Neutron-Capture Patterns and Actinide Production in the I’itoi Substructure with Gemini South/GHOST”

2023

- GS/GHOST | 1.5 hr GS-2023A-SV-101
“High-resolution spectroscopic follow-up of Dynamically Associated Metal-Poor star”

2022

- GN/GRACES | 17.0 hr GN-2022B-Q-107, –306
“Revealing the Chemical Properties of the Progenitor Dwarf Galaxies of Dynamically Associated Metal-Poor Stars”

2021

- GN/GRACES | 14.0 hr GN-2021B-Q-207
“Detailed Chemical Abundance Analysis of Dynamically Associated Metal-Poor Stars”

2020

- GN/GRACES | 10.0 hr GN-2020A-Q-128, –225
“Unveiling the Chemodynamical Evolution of the Milky Way Through Ultra Metal-Poor Stars”

2019

- GN/GRACES | 10.0 hr GN-2019B-Q-115, –219, –310
“Investigation of the Nature of Population III Stars by Ultra Metal-Poor Stars ($[Fe/H] < -4.0$)”

2018

- GN/GRACES | 10.0 hr GN-2018B-Q-122
“Unveiling the Nature of First-Generation Stars by High-Resolution Spectroscopy with GRACES”

Successful Observing Proposals as Co-I

2026

- GN & GS/GMOS | 9.5 hr GN-2026B-Q-131, GS-2026A-Q-315
“Constraining the Nature of Group IV CEMP Stars”
- GN/IGRINS-2, Gemini-S/GHOST | 6.4 hr GN-2026B-Q-124, GS-2026B-Q-219
“Coordinated Optical and Near-IR Spectroscopy of Planet-Hosting Wide Binaries with Gemini/GHOST and IGRINS-2”
- GN & GS/GMOS | 18.5 hr GN-2026A-Q-128, GS-2026A-Q-124, –320
“Probing the Origins of Group IV CEMP Stars with Gemini/GMOS Medium-Resolution Spectroscopy”
- GN/IGRINS-2 | 6.3 hr GN-2026A-Q-320
“Investigating the Origin of Fluorine in CEMP-no Stars with Gemini-N/IGRINS-2 High-Resolution Spectroscopy”

2025

- LBT/PEPSI | 6.0 hr ND_CEMPG4
“High-resolution Spectroscopy of Group IV CEMP-no Candidates: Testing Early Carbon Enrichment Channels”
- LBT/PEPSI | 3.1 hr ND_RPED
“Chemical Studies of R-process-enhanced Thin-disk Stars”

2024

- LBT/PEPSI | 5.0 hr ND_RPED
“Chemical Studies of R-process-enhanced Thin-disk Stars”
- GS/GMOS | 3.48 hr GN-2024B-FT-209
“Photoevaporating Protoplanetary Disks in Orions’ NGC 1977”
- GN/MAROON-X | 8.0 hr GN-2024B-Q-239
“Transmission Spectroscopy of an Ultra Hot Jupiter TOI-1431b”

2023

- LBT/PEPSI | 10.0 hr ND_ChemPec
“Detailed Chemical Abundances of Chemically Peculiar Stars”

2022

- Subaru/HDS | 10.0 hr S22A-TE008-G
“Identification of r-process Enhanced Metal-poor Stars”

2021

- GN/GMOS | 4.3 hr GN-2021B-Q-204, –303
“Identification of Bright Carbon-Enhanced Metal-Poor Stars in the LAMOST”

2017

- GN/GRACES | 10.0 hr GN-2017B-Q-18
“Investigation of Progenitors of Ultra Metal-Poor (UMP) Stars by High-Resolution Spectroscopy with GRACES”
- BOAO/BOES | 3.5 nights B02
“Study of EL CVn-type Eclipsing Binaries”

Presentations (Talks/Colloquia/Seminars)

2026

- Aug | 7th K-GMT Science Program Users’ Meeting, Daejeon, Korea
“Neutron-capture Signatures of Accreted Halo Building Blocks from Gemini Spectroscopy of Metal-Poor Stars”
- Aug | NOIRLab North Colloquium, Hilo, Hawaii, USA
“Chemical Fossils of the Milky Way: From GRACES to GHOST”
- Apr | 113rd KAS Spring Meeting, Suwon, Korea
“Uncovering Extremely Metal-Poor Stars in the Southern Hemisphere: from 7DS to Gemini”
- Mar | GHOST Webinar I, Korea Gemini Office at KASI (online)
“Science Opportunities with Gemini/GHOST: High-Resolution Spectroscopy Across Astrophysics”
- Jan | 3rd SMILE Workshop, Seoul, Korea
“An r-process-enhanced Metal-poor Star Associated with a Milky Way Halo Substructure”
- Jan | Galaxy Forum in Yonsei University, Seoul, Korea
“The Memory of Metals: Tracing Galactic Events through Stellar Chemistry”

2025

- Oct | 112th KAS Fall Meeting, Jeaju, Korea
“The Chemical Fingerprints of the Early Milky Way”
- Sep | 2025 China-Japan-South Korea International Workshop on the Milky Way and Exoplanets, Hangzhou, China
“R-process Enrichment and Actinide Boost in Very Metal-Poor Star LAMOST J122216-063345”

2024

- Aug | K-GMT Program Users Meeting 2024, Daejeon, Korea
“Chemodynamical Analysis of Dynamically Associated Metal-Poor (DAMP) Star with GEMINI-S/GHOST”
- May | DREAM meeting, Daejeon, Korea
“Search for Extremely Metal-Poor ($[Fe/H] < -3.0$) Stars in the Milky Way”

2023

- Aug | Noirlab South Colloquium, La Serena, Chile
“Search for Extremely Metal-Poor ($[Fe/H] < -3.0$) Stars with GEMINI”
- Feb | Stars and Milky Way Workshop, Yeosu, Korea
“Search for Extremely Metal-Poor ($[Fe/H] < -3.0$) Stars with GEMINI/GRACES”

2022

- Oct | 106th KAS Fall Meeting, Special Session: K-GMT Science Program Users Meeting 2022, Gyeongju, Korea
“High-resolution optical spectroscopic observation of Extremely Metal-Poor ($[Fe/H] < -3.0$) Stars with GEMINI/GRACES”

2021

- Dec | 2021 High-Resolution Spectroscopy and GMT Research Topic Development Mini Workshop, Jeonju, Korea
“High-resolution Spectroscopy of Very Metal-Poor ($[Fe/H] < -2.0$) Stars with GEMINI/GRACES”

2019

- Oct | 101th KAS Fall Meeting, Seoul, Korea
“Chemodynamics of Ultra Metal-Poor Stars in the Milky Way”
- Aug | Brain Korea plus 21 Workshop, Gochang, Korea
“Investigation of Nature of First-Generation Stars By Ultra Metal-Poor ($[Fe/H] < -4.0$) Stars”
- Jul | 2nd Milky Way Mini Workshop, Daejeon, Korea

“Investigation of Nature of First-Generation Stars By Ultra Metal-Poor ($[\text{Fe}/\text{H}] < -4.0$) Stars”

2018

- Jul | 1st Milky Way Mini Workshop, Seoul, Korea
“High-resolution spectroscopy of Ultra Metal-Poor ($[\text{Fe}/\text{H}] < -4.0$) Candidates”

Presentations (Posters)

2025

- Jun | International Symposium on Nuclei in the Cosmos XVIII (NIC XVIII), Girona, Spain
“R-process Enrichment and Actinide Boost in Very Metal-Poor Star LAMOST J122216-063345”

2024

- Jun | ELT Science in Light of JWST, Sendai, Japan
“Chemodynamical Analysis of Metal-Poor Stars in the Milky Way”

2022

- Aug | IAUGA 2022 XXXIst General Assembly International Astronomical Union, Busan, Korea
“Unveiling the Nature of Progenitors of Known Substructures in the Galactic Halo by Their Metal-Poor Member Stars”
- Jul | Gemini Observatory Science Meeting 2022, Seoul, Korea
“Chemodynamical Properties of Extremely Metal-Poor Stars Observed with Gemini/GRACES”

2020

- Jan | 235th AAS Meeting, Hawaii, USA
“Chemodynamical Properties of Extremely Metal-Poor Stars”

2019

- Apr | 100th KAS Spring Meeting, Busan, Korea
“Early Chemical Evolution of the Milky Way Revealed by Ultra Metal-Poor ($[\text{Fe}/\text{H}] < -4.0$) Stars”

2018

- Dec | Stellar Archaeology as a Time Machine to the First Stars, Tokyo, Japan
“Investigation of the Nature of First-Generation Stars with Ultra Metal-Poor ($[\text{Fe}/\text{H}] < -4.0$) Stars”
- Feb | K-GMT Science Program Users Meeting 2018, Daejeon, Korea
“The Nature of First-Generation Stars as Revealed by Ultra Metal-Poor ($[\text{Fe}/\text{H}] < -4.0$) Stars”

2017

- Dec | A Celebration of CEMP & Gaia of GALAH, Melbourne, Australia
“The Nature of First-Generation Stars as Revealed by Ultra Metal-Poor ($[\text{Fe}/\text{H}] < -4.0$) Stars”

Teaching and Invited Lectures

Invited Guest Lectures

- “All About Observing with Gemini” Apr 2026
– Observational Astronomy, Yonsei University

Teaching Assistant

- Programming Language and Practice (Python) Spring 2021
- Wisdom of Astronomy Spring 2020, Fall 2019
- Galactic Astronomy and Practice I & II Spring & Fall 2019
- Programming Language and Practice (IDL) Spring 2019, Spring 2018
- Astrophotography and Practice Spring & Fall 2019, 2017
- Mechanics II and Exercise Fall 2017

Other Experience

Leadership and Organizational Roles

- President of Graduate Student Association, CNU Jan–Dec 2021
- President of Astronomy and Space Science Department Student Association, CNU Jan–Dec 2014
- Representative of Female Students, College of Natural Sciences in CNU Mar–Dec 2013

Community Service and Ethical Engagement

- Member of Women’s Division, Korean Astronomical Society Aug 2017–Present
- Member of University Human Rights Committee, CNU Jan–Dec 2021
- Certified as a Second-Class Amateur Astronomy Instructor, Korean Amateur Astronomical Society, Korea Apr–Dec 2011

Outreach and Mentorship

Outreach

- Guest Instructor, Science Education Institute for the Gifted, Kongju National University Jun 2026
– “*Tracing the Milky Way’s Past through Stellar Chemical Abundances and Kinematics*”
- Geoscience Program Assistant, Gifted Education Center of CNU, Korea Mar 2018–Aug 2019
- Planetarium Storytelling Narrator, National Science Museum, Korea Aug 2011–Aug 2014
- Assistant Observing Evaluator at the Student Astronomical Observation Competition, Korea Oct 2011 & 2012

Mentorship

- Invited mentoring talk, KYAM Winter Workshop 2026 Jan 2026
– “*Navigating Graduate School: Perspectives on Research and Academic Life*”
- Mentor for Undergraduate Capstone Project, CNU, Korea Spring 2020, Fall 2021, Fall 2024
- Organized and Conducted a Linux + IRAF Data Processing Seminar at Butterstar Astronomy Club in Dongducheon High School, Korea Feb 3–5, 2022
- Mentor for SW Arduino training for elementary school students, Software-Centered University Program in CNU, Korea Jan 22, 2019
- Mentor for Prospective Female Astronomy Students, College of Natural Science in CNU, Korea Aug 2012

Media

- Interviewed for the ‘Easy Universe’ column in *Top Student Science magazine* Oct 2023
- Authored an opinion piece titled “Answering the Mysteries of the Universe” for *Kyosu Shinmun (Newspaper)* Mar 2022

Technical skills

Programming Languages	Python, IDL, R, $\text{\LaTeX}$, Shell scripting
Environments	Unix, Linux (Centos, Redhat, Ubuntu), Mac, Windows
Tools	IRAF, MOOG, SPECTRE, ARES, DRAGRACES, DRAGONS, SMH

Publications

First Author

1. **Jeong**, et al., “Chemodynamical Investigation of Very Metal-Poor Stars in the Milky Way Substructures with Gemini-N/GRACES”, in preparation.
2. **Jeong**, et al., “The Actinide-Boost Star LAMOST J122216.85-063345.2: A Detailed *R*-process Abundance Study with Gemini-S/GHOST”, 2026, *ApJ*, 1005, 92.
3. **Jeong**, Lee, Beers, Placco et al., “Search for Extremely Metal-poor Stars with Gemini-N/Graces. I. Chemical-abundance Analysis”, 2023, *ApJ*, 948, 38.

Contributing Author

1. Lee et al. incl. **Jeong**, “A New Subclass of Carbon-enhanced Metal-poor Stars at Extremely Low Metallicity”, 2025, 2025, *ApJL*, 991, L42.
2. Rodriguez et al. incl. **Jeong**, “The Secondary Component of the Extremely Low Mass Ratio Massive Binary HD 165246 Revealed by GHOST”, 2025, 2025, *ApJ*, 980, 169R.
3. Kalari et al. incl. **Jeong**, “Gemini High-resolution Optical SpecTrograph (GHOST) at Gemini-South: Instrument performance and integration, first science, and next steps”, 2024, 2024, *AJ*, 168, 208.

4. Deibert et al. incl. **Jeong**, “High-resolution Dayside Spectroscopy of WASP-189b: Detection of Iron during the GHOST/Gemini South System Verification Run”, 2024, 2024, *AJ*, 168, 148D.
5. Placco et al. incl. **Jeong**, “SPLUS J142445.34-254247.1: An r-process-enhanced, Actinide-boost, Extremely Metal-poor Star Observed with GHOST”, 2023, *ApJ*, 959, 60.
6. Jang et al. incl. **Jeong**, “GRACES Observations of Mg-Enhanced Metal-Poor Stars in the Milky Way”, 2023, *JKAS*, 56, 11.

Research notes and Proceedings

1. Heo & **Jeong**, “Discovery of Raman-Scattered C II Lines in V366 Carinae with GHOST/Gemini South”, 2024, *Research Notes of the AAS*, Volume 8, Issue 12, id.304
2. **Jeong**, Lee, Beers, Placco, “Chemodynamical Properties of Extremely Metal-Poor Stars”, 2020, *American Astronomical Society meeting #235*, id. 214.02. *Bulletin of the American Astronomical Society*, Vol. 52, No. 1.